

Point Membership for Robust-to-Linear-Dynamics Linear Programs and Nonnegativity of Linear Recurrences

Draft note

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Abstract

Ahmadi and Günlük's robust-to-linear-dynamics linear programs lead to the following point-membership question: given rational data $A \in \mathbb{Q}^{m \times n}$, $b \in \mathbb{Q}^m$, $G \in \mathbb{Q}^{n \times n}$, and $z \in \mathbb{Q}^n$, decide whether

$$AG^k z \leq b \quad \text{for every } k \geq 0.$$

This note proves an exact effective equivalence between this membership problem and the nonnegativity problem for rational linear recurrence sequences. Thus the result is partial progress rather than a full solution of the general decidability question: a full decision procedure for the membership problem would amount to a full decision procedure for linear-recurrence nonnegativity. The reduction is sharp in the sense that one homogeneous halfspace already captures the full linear-recurrence nonnegativity problem. Known low-order results for linear recurrence sequences nevertheless give unconditional decidability for ambient dimension $n \leq 4$, and for $n \leq 5$ when all right-hand sides are zero.

Status of the result

The result below is *not* an unconditional solution of the full membership problem from robust-to-linear-dynamics linear programming. It is a precise reformulation and hardness result: the membership problem is decidable if and only if rational linear-recurrence nonnegativity is decidable. Since the latter is a classical open decision problem in its unrestricted form, the theorem should be read as partial progress and as an identification of the exact obstruction. A separate, unconditional consequence is obtained in low dimension by invoking the known decidability of nonnegativity for linear recurrence sequences of order at most five.

1 The membership problem

Let

$$P = \{x \in \mathbb{R}^n : Ax \leq b\}, \quad A \in \mathbb{Q}^{m \times n}, \quad b \in \mathbb{Q}^m,$$

and let the discrete-time dynamics be

$$x_{k+1} = Gx_k, \quad G \in \mathbb{Q}^{n \times n}.$$

The invariant feasible set associated with the linear dynamics is

$$\mathcal{S} = \bigcap_{k=0}^{\infty} \{x \in \mathbb{R}^n : AG^k x \leq b\}.$$

The decision problem considered here is the following.

Definition 1 (Robust-to-linear-dynamics membership). *An instance of RLD-MEMBERSHIP consists of rational matrices and vectors (A, b, G, z) , with $A \in \mathbb{Q}^{m \times n}$, $b \in \mathbb{Q}^m$, $G \in \mathbb{Q}^{n \times n}$, and $z \in \mathbb{Q}^n$. The question is whether*

$$z \in \mathcal{S}, \quad \text{equivalently} \quad AG^k z \leq b \text{ for every } k \geq 0.$$

If $a_i^\top$ denotes the i -th row of A , this condition is

$$a_i^\top G^k z \leq b_i \quad (i = 1, \dots, m, k \geq 0).$$

The objective vector of the corresponding linear program is irrelevant for this point-membership question. The problem above is therefore a decision subproblem of the full robust-to-linear-dynamics optimization problem, not a replacement for that optimization problem.

2 Linear recurrence preliminaries

Throughout the note $\mathbb{N} = \{0, 1, 2, \dots\}$.

Definition 2 (Linear recurrence sequence). *A rational linear recurrence sequence, or rational LRS, is a sequence $(u_k)_{k \geq 0}$ of rational numbers admitting a finite-dimensional rational linear representation*

$$u_k = p^\top M^k q \quad (k \geq 0)$$

for some $d \geq 1$, $M \in \mathbb{Q}^{d \times d}$, and $p, q \in \mathbb{Q}^d$. Equivalently, (u_k) satisfies a homogeneous linear recurrence with constant rational coefficients; equivalently, its ordinary generating function is rational.

The decision problem LRS-NONNEGATIVITY asks, for a given rational LRS, whether

$$u_k \geq 0 \quad \text{for every } k \geq 0.$$

The equivalence of the matrix, recurrence, and rational-generating-function presentations is standard and effective. For example, a recurrence of order d ,

$$u_{k+d} = c_{d-1}u_{k+d-1} + \dots + c_1u_{k+1} + c_0u_k,$$

is represented by the companion matrix

$$C = \begin{pmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \\ c_0 & c_1 & c_2 & \dots & c_{d-1} \end{pmatrix}, \quad x_0 = (u_0, u_1, \dots, u_{d-1})^\top,$$

since then $u_k = e_1^\top C^k x_0$. Conversely, by Cayley-Hamilton every sequence $p^\top M^k q$ satisfies a recurrence of order at most d . Its generating function is

$$\sum_{k \geq 0} p^\top M^k q t^k = p^\top (I - tM)^{-1} q,$$

which is rational.

Remark 3 (Conventions about zero trailing coefficients). *Some references require the last recurrence coefficient to be nonzero. This convention does not affect any result below. If the recurrence has a zero root, one checks a finite initial segment and then applies the recurrence with the zero root removed to the remaining tail. Equivalently, the matrix and rational function definitions above absorb this harmless convention issue automatically.*

We shall use two elementary closure facts.

Lemma 4 (Affine matrix-orbit coordinates are LRS). *Let $G \in \mathbb{Q}^{n \times n}$, $p, q \in \mathbb{Q}^n$, and $\beta \in \mathbb{Q}$. Then*

$$v_k = \beta - p^\top G^k q$$

is a rational LRS. It has a rational linear representation of dimension at most $n + 1$. If $\beta = 0$, it has one of dimension at most n .

Proof. For the general case, set

$$\widetilde{M} = \text{diag}(1, G), \quad \widetilde{p} = (\beta, -p^\top)^\top, \quad \widetilde{q} = (1, q^\top)^\top.$$

Then

$$\widetilde{p}^\top \widetilde{M}^k \widetilde{q} = \beta - p^\top G^k q.$$

If $\beta = 0$, the representation $(-p)^\top G^k q$ has dimension at most n . □

Lemma 5 (Interleaving finitely many LRS). *Given rational LRS $v^{(1)}, \dots, v^{(m)}$, one can effectively construct a single rational LRS w such that*

$$w_{mj+i} = v_j^{(i+1)} \quad (j \geq 0, i = 0, \dots, m-1).$$

Consequently,

$$w_k \geq 0 \text{ for all } k \iff v_j^{(i)} \geq 0 \text{ for all } i, j.$$

Proof. Let

$$V_i(t) = \sum_{j \geq 0} v_j^{(i)} t^j$$

be the ordinary generating function of $v^{(i)}$. Each $V_i(t)$ is rational and can be computed from any standard finite encoding of $v^{(i)}$. Define

$$W(t) = \sum_{i=0}^{m-1} t^i V_{i+1}(t^m).$$

Then $W(t)$ is rational, hence it is the generating function of a rational LRS. Its coefficient of t^{mj+i} is exactly $v_j^{(i+1)}$. The nonnegativity equivalence is immediate from this coefficient identity. □

3 Exact equivalence with LRS nonnegativity

Let $\text{RLD-MEMBERSHIP}_{1,0}$ denote the special case of RLD-MEMBERSHIP in which there is only one inequality and the right-hand side is zero; that is, $P = \{x \in \mathbb{R}^n : a^\top x \leq 0\}$ for a rational row vector $a^\top$.

Theorem 6 (Effective equivalence). *The decision problem RLD-MEMBERSHIP over rational data is effectively equivalent to LRS-NONNEGATIVITY . More precisely:*

- (a) *Every RLD-MEMBERSHIP instance can be transformed effectively into a single rational LRS w such that*

$$z \in \mathcal{S} \iff w_k \geq 0 \text{ for every } k \geq 0.$$

- (b) *Every rational LRS nonnegativity instance can be transformed effectively into an instance of RLD-MEMBERSHIP_{1,0}. Hence the single homogeneous halfspace case already contains arbitrary rational LRS nonnegativity.*

Proof. For (a), let (A, b, G, z) be an instance of RLD-MEMBERSHIP. Write the rows of A as $a_1^\top, \dots, a_m^\top$, and define, for each row,

$$v_k^{(i)} = b_i - a_i^\top G^k z \quad (k \geq 0).$$

By Lemma 4, every $v^{(i)}$ is a rational LRS. Moreover,

$$AG^k z \leq b \text{ for every } k \geq 0 \iff v_k^{(i)} \geq 0 \text{ for every } i = 1, \dots, m \text{ and every } k \geq 0.$$

Lemma 5 effectively combines these finitely many LRS nonnegativity conditions into the nonnegativity of one rational LRS w .

For (b), let $(u_k)_{k \geq 0}$ be a rational LRS. Using a companion-matrix or linear-representation encoding, write

$$u_k = p^\top M^k q \quad (k \geq 0)$$

with rational M, p, q . Construct a robust-to-dynamics membership instance by setting

$$G = M, \quad z = q, \quad A = -p^\top, \quad b = 0.$$

This instance has one homogeneous halfspace. For every $k \geq 0$,

$$AG^k z \leq b \iff -p^\top M^k q \leq 0 \iff u_k \geq 0.$$

Thus the constructed membership instance is a yes-instance exactly when the original LRS is nonnegative at all indices. $\square$

Corollary 7 (Meaning of the equivalence). *A decision procedure for rational LRS nonnegativity would decide RLD-MEMBERSHIP. Conversely, a decision procedure for RLD-MEMBERSHIP, even restricted to one homogeneous halfspace, would decide rational LRS nonnegativity. Therefore Theorem 6 is not, by itself, a full decidability theorem for RLD-MEMBERSHIP; it identifies LRS-NONNEGATIVITY as the exact remaining barrier.*

4 Consequences

4.1 One halfspace is already hard

The reduction in Theorem 6(b) uses only the halfspace

$$\{x : p^\top x \geq 0\},$$

written in the standard form $-p^\top x \leq 0$. Therefore the possible presence of many facets in the polyhedron P is not the source of the general membership difficulty. The scalar inequality obtained by observing one coordinate, or one rational linear functional, of a linear orbit is already as hard as arbitrary LRS nonnegativity.

4.2 Skolem hardness

Corollary 8 (Skolem hardness). *A decision procedure for RLD-MEMBERSHIP would decide the Skolem problem for integer linear recurrence sequences.*

Proof. Let (u_k) be an integer LRS. The termwise square (u_k^2) is again an LRS: if $u_k = p^\top M^k q$, then

$$u_k^2 = (p \otimes p)^\top (M \otimes M)^k (q \otimes q).$$

Hence

$$r_k = u_k^2 - 1$$

is also an integer LRS. Because each u_k is an integer,

$$r_k \geq 0 \text{ for every } k \iff u_k \neq 0 \text{ for every } k.$$

Thus an algorithm for LRS nonnegativity decides whether an integer LRS has no zero term; complementing the answer decides the Skolem problem. By Theorem 6, any decision procedure for RLD-MEMBERSHIP would give such an algorithm. $\square$

4.3 Low-dimensional decidable cases

We next record the unconditional low-dimensional consequence. The only external input is the known decidability of nonnegativity, often called positivity in the linear-recurrence literature, for LRS of order at most five.

Lemma 9 (Rational order-five instances reduce to integer ones). *For every rational LRS given by a recurrence of order d , one can effectively construct an integer LRS of order at most d with the same sign pattern. In particular, decidability of nonnegativity for integer LRS of order at most five implies decidability of nonnegativity for rational LRS of order at most five.*

Proof. Suppose

$$u_{k+d} = c_{d-1}u_{k+d-1} + \cdots + c_0u_k, \quad c_j \in \mathbb{Q},$$

with rational initial values. Choose a positive integer N divisible by the denominators of all c_j , and then choose a positive integer D such that $DN^i u_i \in \mathbb{Z}$ for $i = 0, \dots, d-1$. Define

$$v_k = DN^k u_k.$$

Then u_k and v_k have the same sign for every k . Moreover,

$$v_{k+d} = \sum_{j=0}^{d-1} c_j N^{d-j} v_{k+j},$$

and each coefficient $c_j N^{d-j}$ is an integer. Hence (v_k) is an integer LRS of order at most d with the same sign pattern. $\square$

Corollary 10 (Low-dimensional membership). *The membership problem RLD-MEMBERSHIP is decidable for all rational instances in ambient dimension $n \leq 4$, with an arbitrary number of inequalities. If $b = 0$, equivalently if the constraint set is a rational polyhedral cone, then RLD-MEMBERSHIP is decidable for all rational instances in ambient dimension $n \leq 5$.*

Proof. For a general instance, membership is equivalent to the simultaneous nonnegativity of the row sequences

$$v_k^{(i)} = b_i - a_i^\top G^k z \quad (i = 1, \dots, m).$$

By Lemma 4, each $v^{(i)}$ has a rational linear representation of dimension at most $n + 1$, hence it satisfies a recurrence of order at most $n + 1$. If $n \leq 4$, this order is at most five. Applying the known order-five nonnegativity decision procedure to each row sequence and accepting only if all row sequences are nonnegative decides membership.

If $b = 0$, then

$$v_k^{(i)} = -a_i^\top G^k z,$$

which has a rational linear representation of dimension at most n . For $n \leq 5$, the same order-five decision procedure applies row by row. $\square$

Remark 11. *The proof of Corollary 10 deliberately decides the row sequences separately. Interleaving the row sequences into one LRS, as in Theorem 6, is useful for exact many-one equivalence, but it can increase the order and would obscure the low-dimensional consequence when the number of inequalities m is large.*

5 Conclusion

The robust-to-linear-dynamics point-membership problem is exactly the nonnegativity problem for rational linear recurrence sequences in another form. The equivalence gives a clean explanation for why the general decidability question is difficult: one homogeneous halfspace already contains the full linear-recurrence nonnegativity problem, and hence also the Skolem problem through the standard square-minus-one reduction. Consequently, the result in this note should be described as partial progress, not as a complete solution. Its unconditional contribution is the exact reduction framework and the resulting low-dimensional decidability consequences, especially dimensions $n \leq 4$ in general and $n \leq 5$ in the homogeneous conic case.

References

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